

Mid-Semester Test
(EC-2 Regular Set 1)

Course No. : AIMLCZG565/ DSECLZG565
Course Title : Machine Learning
Nature of Exam : Closed Book
Weightage : 30%
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No. of Pages	= 4
No. of Questions	= 5

Note to Students:

1. Please follow all the Instructions to Candidates given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1. [5 Marks]

- a) A regression model predicts the monthly electricity bill of a house using four features:

$$\text{Bill} = 120 + 7.5X_1 + 45X_2 - 4X_3 + 18X_4$$

where:

- X_1 = Units consumed in kWh
- X_2 = Number of air conditioners used
- X_3 = Average outdoor temperature in $^{\circ}\text{C}$
- X_4 = Number of family members

- i) Interpret all four coefficients in the regression model. [2 Marks]

- ii) If X_1 is changed from kWh to Wh, explain what happens to the coefficient of X_1 . Will the predicted bill change? [1 Mark]

Note: 1kWh=1000Wh

- b) During gradient descent, the following gradients are observed:

$$\partial J_1 / \partial \theta = 18, \quad \partial J_2 / \partial \theta = 0.02, \quad \partial J_3 / \partial \theta = -12, \quad \partial J_4 / \partial \theta = 0.001,$$

What does the negative gradient value for θ_3 indicate? [1 Mark]

- c) Based only on coefficient magnitudes, which feature appears to have the strongest influence on the electricity bill? Why should this conclusion be treated carefully? [1 Mark]

Q.2. [8 Marks]

A bank wants to build a Decision Tree classifier to predict whether a loan application should be approved based on the following dataset.

Applicant ID	Credit Score	Income Level	Employment Type	Existing Loan	Loan Approved
A1	High	High	Salaried	No	Yes

A2	<u>Low</u>	<u>Low</u>	<u>Self-employed</u>	Yes	No
A3	Medium	High	Salaried	No	Yes
A4	<u>Low</u>	<u>Medium</u>	<u>Unemployed</u>	Yes	No
A5	High	Medium	Self-employed	No	Yes
A6	<u>Medium</u>	<u>Low</u>	<u>Salaried</u>	Yes	No
A7	High	Low	Salaried	No	Yes
A8	<u>Medium</u>	<u>Medium</u>	<u>Self-employed</u>	Yes	No
A9	Low	High	Salaried	No	Yes
A10	<u>Medium</u>	<u>Medium</u>	<u>Unemployed</u>	Yes	No

- Calculate the entropy of the target attribute Loan Approved. [2 Marks]
- Compute the entropy and Information Gain obtained by splitting on the attribute Existing Loan. [2 Marks]
- Construct the complete Decision Tree using the attribute selected as the root node. Show all intermediate branches and leaf nodes. [2 Marks]
- Suppose the bank receives a new application with:
 {Credit Score = High, Income Level = Very High, Employment Type = Contract, Existing Loan = No}
 The values Very High and Contract were not present in the training dataset. Explain how a decision tree can handle such unseen attribute values and recommend an appropriate classification strategy. [2 Marks]

Q3 [8 Marks]

A binary classification model is trained to predict whether a loan applicant will default (1) or not default (0), based on two features - Credit Score and Debt-to-Income Ratio. The raw training data is given below.

Sample	Credit Score (raw)	Debt-to-Income Ratio (raw)	Label y
S1	620	0.80	1
S2	680	0.40	0
S3	590	0.90	1
S4	710	0.30	0

Initial weights: $w_0 = 0.0$, $w_1 = 0.5$, $w_2 = -0.5$. Learning rate $\eta = 0.5$.

(a) Apply Min-Max normalization to the Credit Score column using the formula:

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

- (i) Show the normalized value for each of the four training samples. [1 Mark]
(ii) A new applicant has a raw Credit Score of 650. Compute their normalized x_1 using the same min and max from the training data. [1 Mark]

(b) Use the normalized Credit Score values from Part (a) as x_1 , and treat the Debt-to-Income Ratio column as already normalized for x_2 .

- (i) Compute the predicted probability for each sample using the initial weights. Show the linear combination $z = w_0 + w_1 \cdot x_1 + w_2 \cdot x_2$ and apply the sigmoid function $\sigma(z) = 1 / (1 + e^{(-z)})$ for each sample. [1 mark]
(ii) Compute the Binary Cross-Entropy loss over all four samples: [1 mark]

$$L = -(1/N) \cdot \sum [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$$

(iii) Derive the gradients dL/dw_0 , dL/dw_1 , dL/dw_2 and perform one step of gradient descent to obtain updated weights w_0^* , w_1^* , w_2^* . [2 marks]

(c) The model from Part (b) is overfitting. The team is considering two regularization strategies.

- (i) Your primary goal is to completely eliminate the contribution of irrelevant or noisy features by driving their weights to exactly zero. Should you use L1 (Lasso) or L2 (Ridge) regularization? Explain the key difference in how each penalizes weights and justify why your chosen method achieves this goal. [1 Mark]
(ii) Using the updated weights w_1^* and w_2^* from Part (b)(iii), describe what happens to these weights as regularization strength λ is increased from 0 to a very large value. What is the risk of setting λ too large, and how does this affect the model's predictions on new loan applicants? [1 Mark]

Q4 [6 Marks]

A bank has deployed a binary classifier to detect fraudulent transactions. The positive class ($y = 1$) represents fraud and the negative class ($y = 0$) represents a legitimate transaction. The classifier is evaluated on a test set of 10,000 transactions, of which only 100 are actual fraud cases - making this a highly imbalanced dataset. Two versions of the classifier are evaluated. Their confusion matrices are given below.

Model A:

	Predicted: Fraud (1)	Predicted: Legit (0)
Actual: Fraud (1)	45 (TP)	55 (FN)
Actual: Legit (0)	90 (FP)	9810 (TN)

Model B:

	Predicted: Fraud (1)	Predicted: Legit (0)
Actual: Fraud (1)	80 (TP)	20 (FN)
Actual: Legit (0)	500 (FP)	9400 (TN)

(a) For Model A, compute the following. Show all working.

(i) Accuracy and Error Rate [0.5 Mark]

(ii) Precision, Recall (TPR), False Positive Rate (FPR) and F1-Score [2 Mark]

(iii) Based on the accuracy alone, a colleague concludes that Model A is an excellent classifier. Do you agree? Justify your answer in the context of this fraud detection problem. [1 Mark]

(b) Using the metrics computed above and the values for Model B, answer the following.

(i) Compute the TPR and FPR for both models and plot (or describe) where each model would appear as a point on an ROC curve. [1 Mark]

(ii) Model A has a higher accuracy than Model B. Model B has a higher recall than Model A. Which model would you recommend for the fraud detection use case, and why? (Your answer should reference the cost of False Negatives versus False Positives in this context) [1 Mark]

(iii) What does it mean if a classifier's ROC point lies on the diagonal line? What would the AUC be in that case? [0.5 Mark]

Q5 [3 Marks]

An email service provider is building a binary classifier to detect spam ($y = 1$) versus legitimate email ($y = 0$). Available features include word frequencies, sender domain reputation, email length, punctuation counts, and interaction terms between these features. Two models are trained and evaluated on a 70/30 train-test split.

Model	Description	Training Accuracy	Test Accuracy
A	Logistic Regression using only two features: email length and punctuation count	61%	60%
B	Logistic Regression using all features including all interaction terms (47 features total)	99%	67%

The team also plots learning curves (training and cross-validation accuracy vs. training set size):
Model A: both curves converge and plateau at around 61%.

Model B: training accuracy stays near 99%; cross-validation accuracy rises slowly, reaching only 70% even with the full dataset.

(a) Diagnose each model in terms of bias and variance using the accuracy numbers. Also interpret what each model's learning curve pattern reveals and whether adding more training data would help. [1 Mark]

(b) Model B is overfitting. State any two strategies (without using regularization) to fix this, and briefly explain the mechanism behind each. [1 Mark]

(c) Propose a single improved model that avoids the failure of both Model A and Model B. Explain how it balances bias and variance. [1 mark]
